

# ATTUNE — Software Overview

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*A technical orientation for someone new to the codebase. Generated from the repository on 2026-06-04.*

**How to read this doc:** every claim here was checked against the actual code in `backend/` and `frontend/`. Where something was **inferred** rather than confirmed from code, it is tagged **[ASSUMPTION]** so you can correct it.

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## 1. Summary

**ATTUNE is a medication-adherence web app — paired with an optional (currently stubbed) physical pill device — that removes the guesswork from *when* to take a medication and makes consistency visible over time.** During a short onboarding, the user names one medication; the backend normalizes the drug via RxNorm/RxNav, pulls the real "dosage & administration" text from the OpenFDA drug-label API, and uses a rule-based extractor to propose a time-of-day window grounded in a quote from that label. That window is resolved to a concrete clock time against the user's own daily routine (wake/sleep/meals), then surfaced as a one-tap "taken / remind me later" home screen, an adherence trend, optional wellness check-ins, a rule-based pattern insight, and an optional caregiver summary. It solves the fact that medication non-adherence is driven as much by *friction and uncertainty* ("when am I even supposed to take this?") as by forgetting — so the product answers the "when," lowers setup friction, and keeps both the patient and the people who support them in the loop.

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## 2. Users & Roles

ATTUNE today is a **single-account application**: one person signs up, onboards, and uses the app. There is **no separate login or invite for a distinct caregiver** — the "caregiver view" is a tab *within the patient's own account*, unlocked by a feature toggle. The roles below are therefore product-level personas served by one technical account type, except where noted.

| Role                          | Who they are                                                                      | What they need                                                                                    | How the code serves them                                                                       | Permissions / gating                                                                                                                                                                                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Patient / primary user</b> | An adult on a new prescription, an elderly patient, or a wellness/supplement user | Get the timing right, log doses with one tap, see their consistency                               | Full app: onboarding, dashboard, schedule settings, logging, insights                          | Authenticated account (email + password, JWT bearer token)                                                                                                                                                                                                                                                       |
| <b>Caregiver (as a view)</b>  | A family member or doctor the patient wants to share with                         | Visibility into adherence %, missed doses, mood/physical averages, an alert on consecutive misses | GET /caregiver/summary + the "Caregiver View" tab                                              | Same account; gated by <code>features.caregiverAccess === true</code> . Returns HTTP 403 ( <code>caregiver_access_disabled</code> ) if the toggle is off<br><b>[ASSUMPTION: caregiver is the same human or a trusted person physically/credential-sharing the patient account — code has no second identity]</b> |
| <b>Developer / operator</b>   | Whoever runs the app locally                                                      | Seed data, drive the API, health checks                                                           | <code>backend/scripts/ ( seed_logs.py , drive_app.py , test_schedule.py ), GET /healthz</code> | Local shell access                                                                                                                                                                                                                                                                                               |

There is **no admin role, no role-based access control, and no multi-tenant separation** beyond "a user can only read/write their own `user_id` -scoped documents."

### 3. Features

Grouped by area; each line reflects code that exists today.

#### Onboarding & profile

- 5-step guided setup: name → medication → when-to-take → daily routine → feature toggles ( `frontend/components/onboarding.tsx` ).
- Browser-detected timezone stored as an IANA zone on the profile.

#### AI / drug-timing suggestion (the differentiator)

- Drug-name normalization via RxNorm/RxNav ( `/REST/rxcui.json` ).

- Real label fetch from OpenFDA ( `/drug/label.json` ).
- Rule-based time-of-day window + grounded "why" extraction, with negation handling ( `backend/app/services/drug_timing.py` ).
- Three confidence tiers: **grounded** (quote from label) / **unverified** (optional LLM, off by default) / **manual** (plain picker).

## Scheduling

- Default weekly schedule (time + days of week).
- Per-weekday overrides and date-range overrides ( `shift` / `set` / `pause` , pause wins).
- Routine model (wake/sleep, with-food + meal times, variable days); changing routine re-derives AI-sourced dose time.
- Conflict detection (outside awake hours, not near a meal when with-food, overlapping overrides).
- Next-due + 7-day upcoming forecast.

## Logging & adherence

- One-tap `taken` / `missed` log, one entry per day (upsert by `date_key` ).
- 30-day adherence trend percentage ( `/stats/trend` ).
- Calendar/heatmap of recent history in the dashboard.

## Wellness check-ins (optional toggle)

- Post-dose modal: physical (1–5), emotional (1–5), optional note ( $\leq 500$  chars), attached to the day's log.

## Insights (rule-based, free)

- Wed/Thu skip-pattern detector with evidence ( `/insights/pattern` ), gated on a minimum log count.

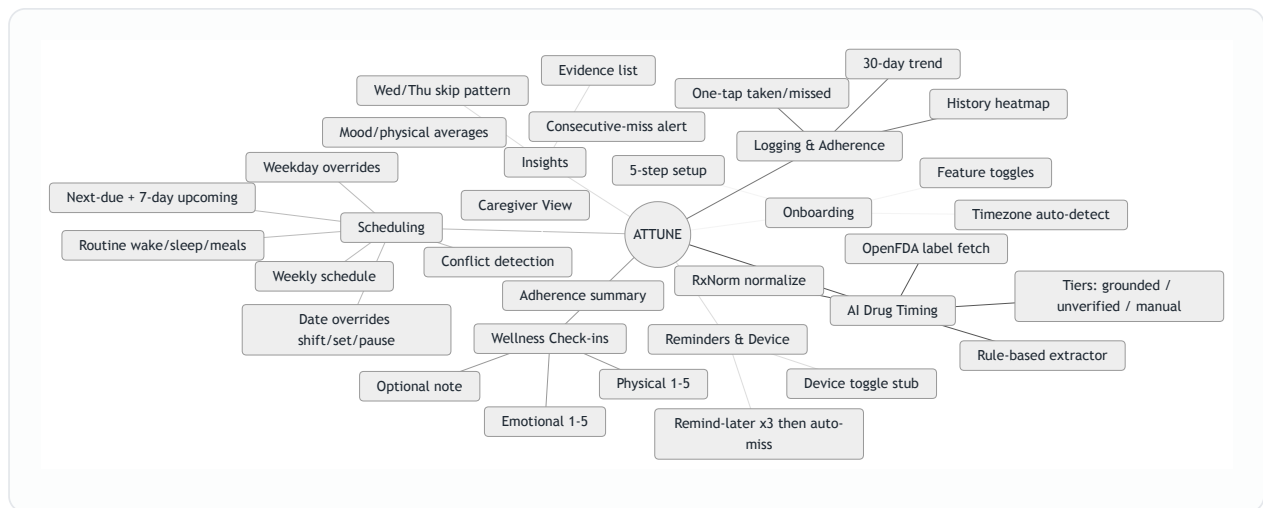
## Caregiver view (optional toggle)

- 30-day summary: adherence %, missed doses, avg physical/mood, recent activity, alert on 2+ consecutive misses.

## Reminders & device

- "Remind me later" counter; auto-logs `missed` after 3 snoozes.
- Device connected/disconnected toggle (a stub standing in for hardware).

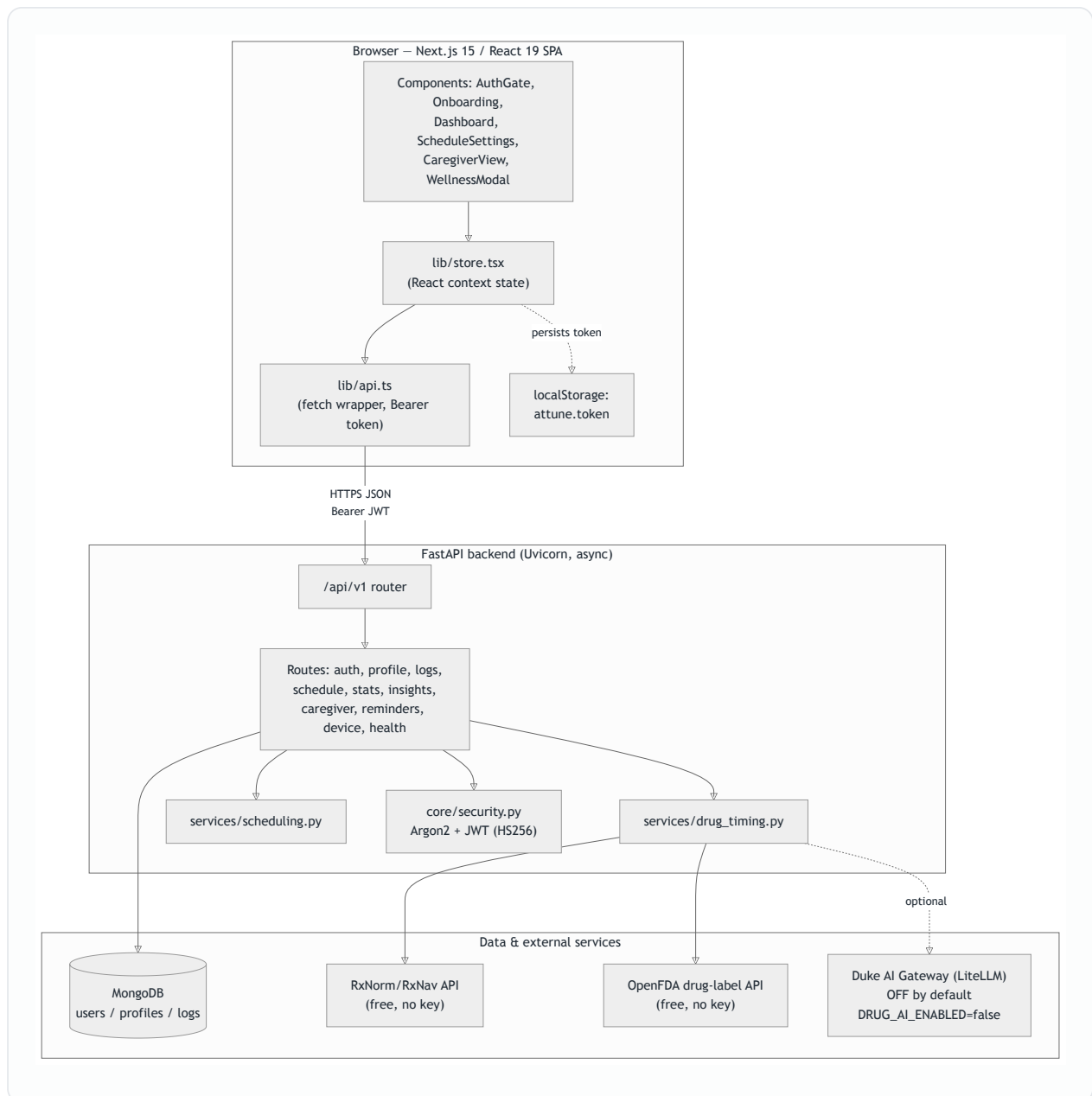
## Diagram — Feature map (Mermaid mindmap)



## 4. Architecture

Two independently-running processes: a **Next.js frontend** (browser SPA) and a **Python/FastAPI backend**, talking over a JSON REST API at `/api/v1`. The backend persists to **MongoDB** and calls two **free external data sources** (RxNorm/RxNav, OpenFDA). An **optional LLM** (Duke AI Gateway via the OpenAI SDK) is wired in but **off by default**.

### Diagram — System architecture (Mermaid flowchart)



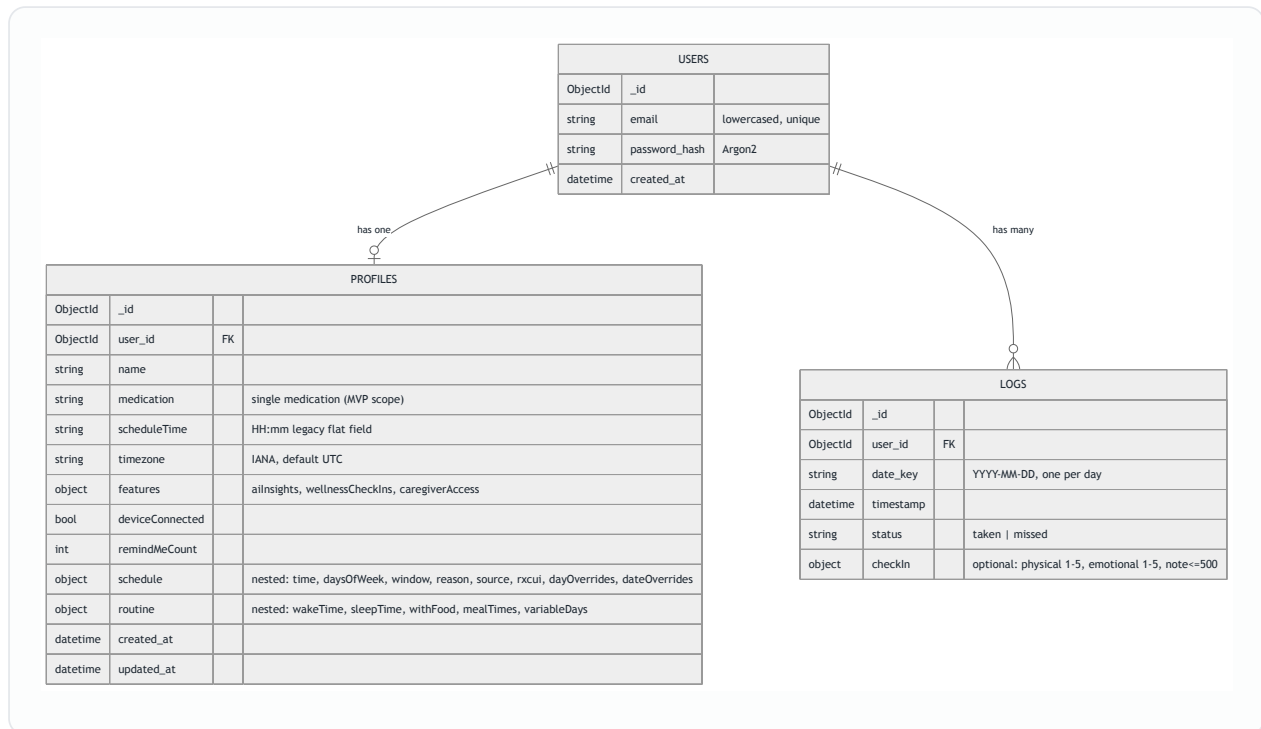
### Key architectural facts (from code):

- Auth is stateless: `create_access_token` (HS256, 7-day expiry) → bearer token → `get_current_user` decodes it and loads the `users` document on every protected request.
- CORS is restricted to `CORS_ORIGINS` (default `http://localhost:3000`).
- `services/scheduling.py` holds **all time math** (window→clock-time resolution, overrides, conflicts, next-due) and is timezone-aware via `ZoneInfo`.
- `services/drug_timing.py` is the only place external drug APIs (and the optional LLM) are called.

## 5. Data Model

Three MongoDB collections. Documents are scoped to a user via `user_id`. The `profiles` document carries two nested sub-documents ( `schedule` , `routine` ); a legacy flat `scheduleTime` field is kept in sync for backward compatibility.

### Diagram — Data model (Mermaid erDiagram)



### Notable details:

- **One log per user per day** — logging upserts on ( `user_id` , `date_key` ) .
- `schedule.source` is "ai" or "user" ; only "ai" -sourced times get re-derived when the routine changes.
- `schedule.dateOverrides[]` entries carry { `id` , `start` , `end` , `type` , `note` } plus `shiftMinutes` (type `shift` ,  $\pm 720$ ) or `time` (type `set` ).
- `features` defaults: `aiInsights=true` , `wellnessCheckIns=true` , `caregiverAccess=false` .

## 6. Data Pipelines

Two pipelines matter most: **(A) the drug-timing suggestion** that runs during onboarding, and **(B) dose logging → adherence/insights** that runs daily.

Diagram — Pipeline A: drug-timing suggestion (Mermaid flowchart)

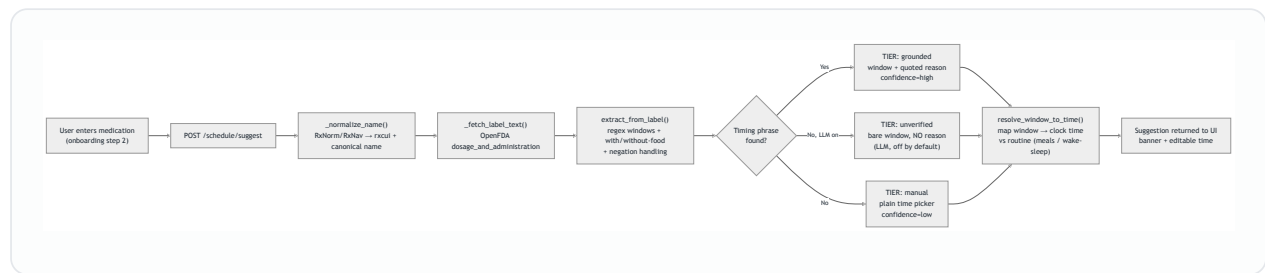
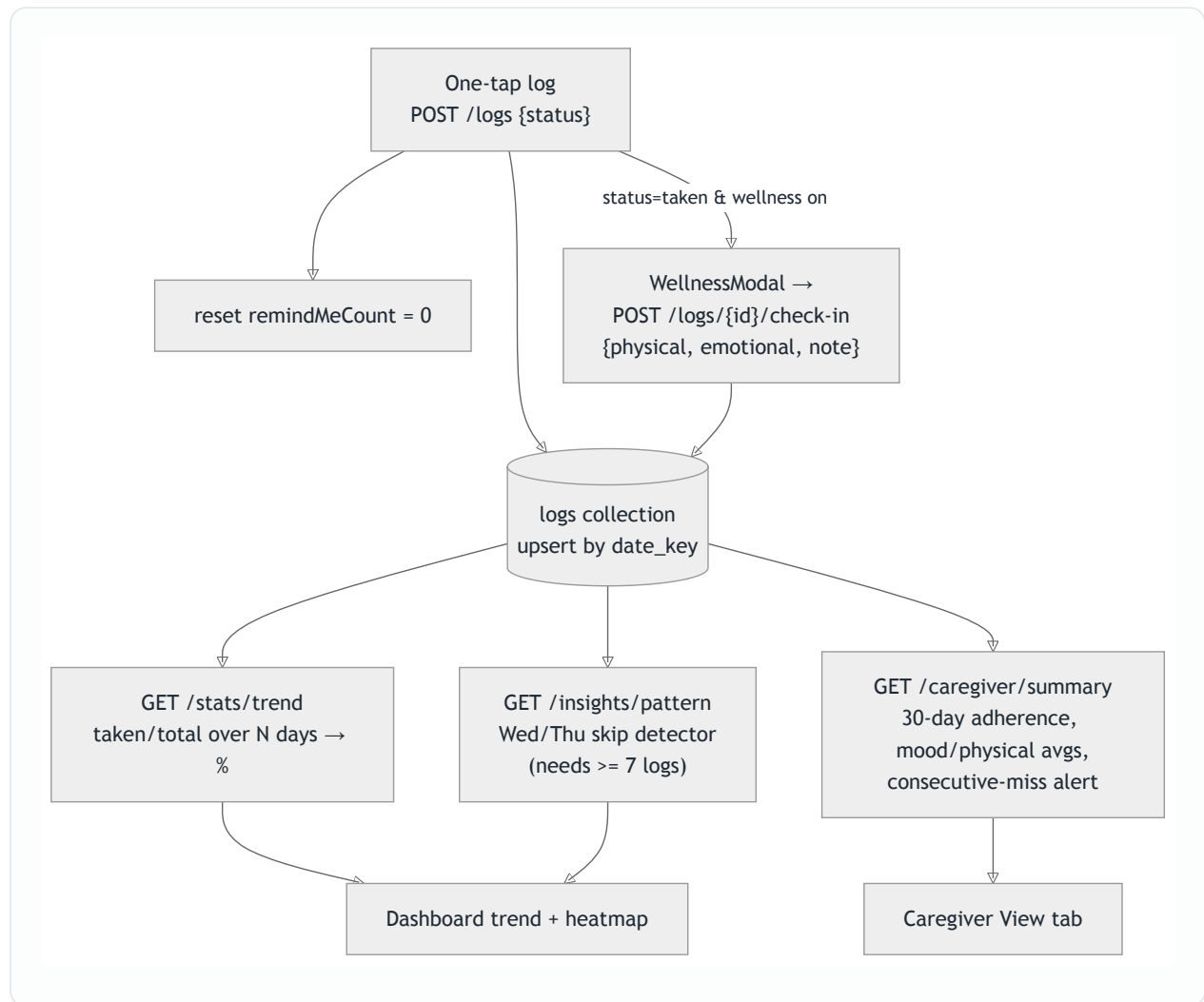


Diagram — Pipeline B: logging → adherence, insights, caregiver (Mermaid flowchart)



## 7. Use of AI

### Where AI is (and isn't)

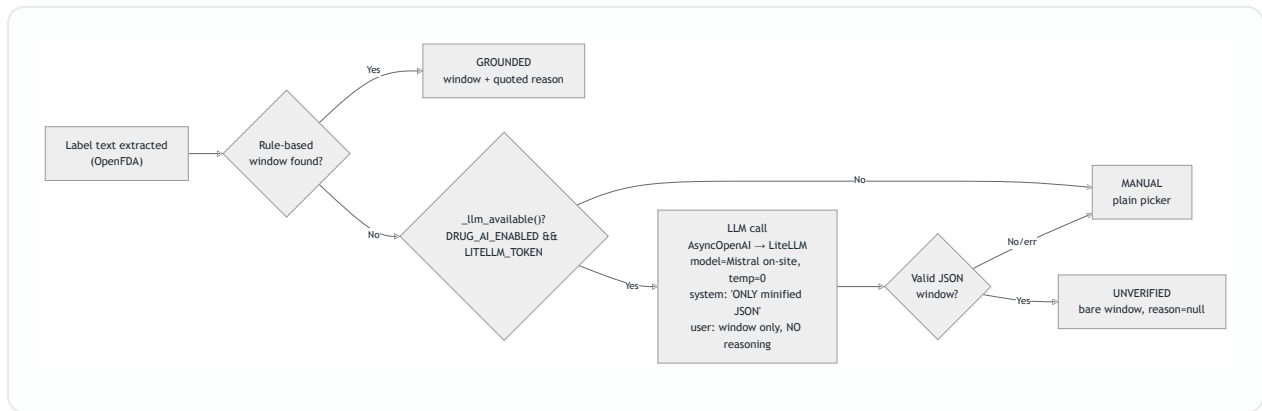
| Surface                             | Technique                                                                                                                                                | Model                                                                                  | Default                                              | Status                                                      |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------|
| <b>Drug-timing "why" extraction</b> | Rule-based regex over real FDA label text, with negation filtering                                                                                       | None (deterministic code)                                                              | Always on                                            | Built; this is the primary "AI" experience                  |
| <b>Unverified timing fallback</b>   | LLM completion (OpenAI-compatible chat) constrained to return <i>only</i> a window, JSON, temperature 0, <b>forbidden from inventing a justification</b> | Duke AI Gateway / LiteLLM, <code>LITELLM_MODEL</code> default "Mistral on-site" (free) | <b>OFF</b><br>( <code>DRUG_AI_ENABLED=false</code> ) | Built but gated; only fires when label has no usable timing |
| <b>Pattern insight</b>              | Rule-based statistical detector (Wed/Thu misses in last 8 occurrences, $\geq 4$ threshold)                                                               | None                                                                                   | On if <code>aiInsights</code> toggle                 | Built (no LLM)                                              |

**Why AI/this design fits here:** the product's hardest job is answering "when should I take this?" *trustworthily*. A free-text medical chatbot would risk inventing advice, so the design deliberately makes the **rule-based, label-grounded extractor the default** and treats the LLM as an optional, justification-free fallback. This matches the project's guardrail that timing suggestions **must be free and grounded, and never surface an ungrounded "why."** The "pattern insight" is branded as AI but is a transparent, free heuristic — chosen because it is explainable and cannot hallucinate.

**[ASSUMPTION]** The label "AI" in the product is largely marketing for a rule-based pipeline plus an optional LLM. The code confirms the LLM path exists; I have **not** confirmed it has ever been exercised with a live `LITELLM_TOKEN` , since the flag defaults off.

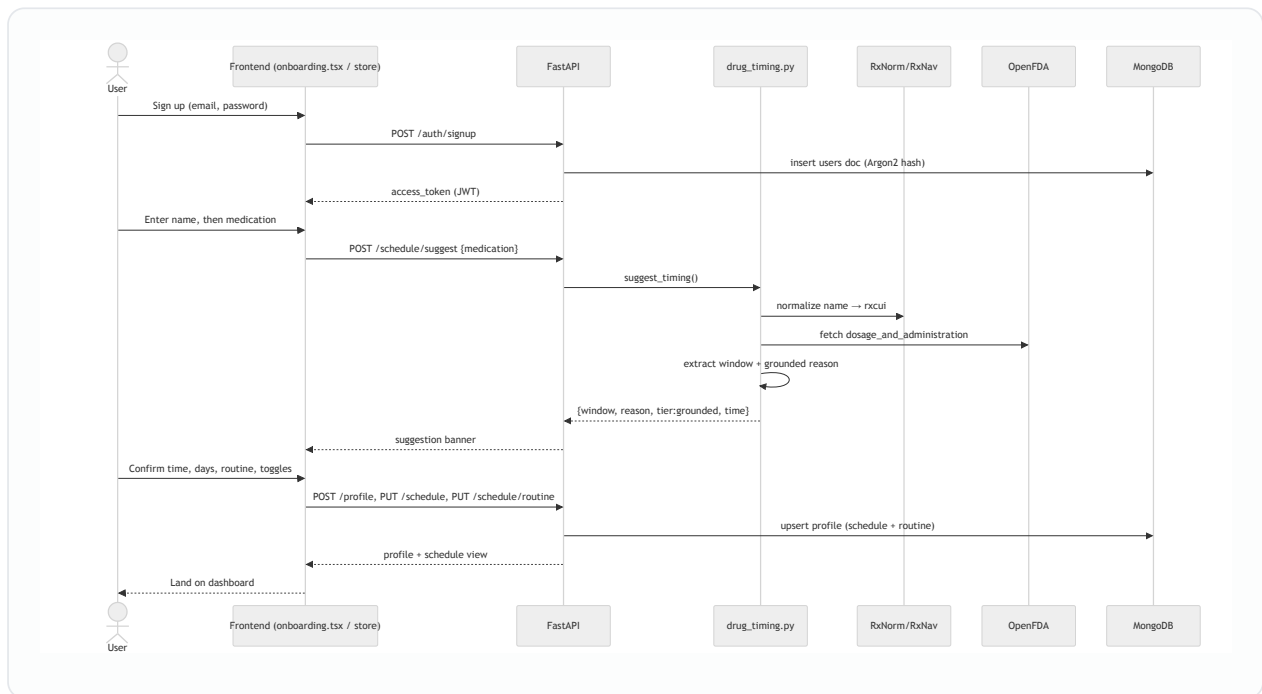
### Diagram — The AI/LLM step inside the timing pipeline (Mermaid flowchart)





## 8. Key User Flows

### Diagram — Flow 1: Onboarding with grounded timing suggestion (Mermaid sequence)



### Diagram — Flow 2: Daily dose log + wellness check-in (Mermaid sequence)



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## 9. PRD Field Extract

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This table reports only **what the repository actually shows** for each field — it makes no comparison to any PRD and no judgement about whether the code matches a spec.

| PRD Field                | What the Repo Shows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Overview</b>  | A medication-adherence web app that looks up <i>when</i> to take one medication from the real FDA label, fits it to the user's routine, and provides one-tap logging, an adherence trend, optional wellness check-ins, a rule-based pattern insight, and an optional caregiver summary. (Confirmed by <code>product-skeleton.md</code> and the implemented routes/services.)                                                                                                                                                              |
| <b>Who it's for</b>      | A single authenticated patient/primary user (adults on a new prescription, elderly patients, or wellness/supplement users). A "caregiver" is served as a <i>view within the same account</i> gated by <code>features.caregiverAccess</code> , not a separate identity. No admin/role system in code.                                                                                                                                                                                                                                      |
| <b>Priority Features</b> | Implemented in code: email+password auth (JWT/Argon2); 5-step onboarding; drug-timing suggestion (RxNorm + OpenFDA + rule-based, 3 tiers); routine-aware schedule with weekday + date overrides and conflict detection; next-due + 7-day upcoming; one-tap taken/missed logging; 30-day trend; Wed/Thu pattern insight; wellness check-ins; caregiver summary with consecutive-miss alert; remind-later→auto-miss; device toggle stub.                                                                                                    |
| <b>Core Workflow</b>     | <b>Trigger:</b> a scheduled dose is due (dashboard shows "next dose" + reason). <b>Action:</b> user taps "Log as Taken" (or "Remind me later"), optionally completes a wellness check-in. <b>Result:</b> the day's log is upserted, <code>remindMeCount</code> resets, and the trend/heatmap/insight/caregiver summary update.                                                                                                                                                                                                            |
| <b>Aha Moment</b>        | During onboarding step 3: the user types one medication and ATTUNE returns a <b>grounded</b> timing suggestion — a window <i>plus a real sentence quoted from the FDA label</i> — resolved to a concrete time against their routine, instead of a blank time field. ( <code>POST /schedule/suggest</code> → <code>tier:"grounded"</code> .)                                                                                                                                                                                               |
| <b>Inputs</b>            | Email + password; user's name; one medication name; chosen window/exact time + days of week; routine (wake/sleep, with-food + meal times, variable days); browser-detected IANA timezone; feature toggles; dose status ( <code>taken</code> / <code>missed</code> ); wellness check-in (physical 1–5, emotional 1–5, note ≤500 chars); schedule overrides (weekday, date-range shift/set/pause).                                                                                                                                          |
| <b>Outputs</b>           | A grounded/unverified/manual timing suggestion; a resolved weekly schedule with next-due + 7-day upcoming + conflict warnings; adherence trend percentage; recent-history heatmap; a Wed/Thu pattern message with evidence; a caregiver summary (adherence %, missed count, avg physical/mood, recent activity, consecutive-miss alert); device connected/disconnected flag.                                                                                                                                                              |
| <b>Use of AI</b>         | (1) Default path is <b>rule-based</b> (no model): regex extraction of a timing window + quoted reason from FDA label text, with negation handling. (2) Optional <b>LLM</b> via Duke AI Gateway / LiteLLM (OpenAI SDK, <code>AsyncOpenAI</code> , <code>LITELLM_MODEL</code> default "Mistral on-site" , temp 0), gated behind <code>DRUG_AI_ENABLED=false</code> , constrained to return a bare window with <code>reason:null</code> and forbidden from inventing justification. (3) Pattern insight is a rule-based heuristic, no model. |

| PRD Field                        | What the Repo Shows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MVP Scope</b>                 | One medication per user. Fully implemented backend (25 endpoints across auth/profile/logs/schedule/stats/insights/caregiver/reminders/device/health) + Next.js frontend (onboarding, dashboard, schedule settings, caregiver view, wellness modal). Caregiver is a same-account view. Device is a boolean toggle. LLM path present but off.                                                                                                                                                                                                                                                                   |
| <b>Constraints / Must Not Do</b> | Single medication only (no multi-med/polypharmacy/interaction checks — RxNav interaction API noted as discontinued). Timing "why" must be <b>grounded</b> (quoted from label); the unverified LLM tier must <b>not</b> surface a justification ( <code>reason:null</code> ). LLM is off by default and uses the free model to avoid budget. API key never hardcoded/committed (lives in gitignored <code>.env</code> ). One log per user per day. Auth required on all endpoints except <code>/healthz</code> , <code>/auth/signup</code> , <code>/auth/login</code> . CORS restricted to configured origins. |

## 10. Assumptions & Open Questions

### Assumptions made while documenting (correct me if wrong):

- **[ASSUMPTION]** The "caregiver" is either the patient themselves reviewing their data or a trusted person using the patient's credentials — the code has no second identity, invite, or separate caregiver login.
- **[ASSUMPTION]** The optional LLM path has not necessarily been exercised against a live Duke Gateway key, since `DRUG_AI_ENABLED` defaults to `false` ; I confirmed the code path exists, not that it has run in production.
- **[ASSUMPTION]** The physical pill device is not integrated at all — `device.py` is a pure boolean toggle with no hardware/BLE/push code, consistent with the skeleton calling it a stub.
- **[ASSUMPTION]** `httpx` is the async HTTP client used by `drug_timing.py` (the import is used in code) even though it is not pinned in `requirements.txt` ; it appears to be relied on as an available dependency.

### Open questions surfaced (some flagged in `product-skeleton.md` ):

- Post-setup landing: should the user land on the calendar, the next-dose home, or the first scheduled dose? (Marked in-progress.)
- Can a second medication be added post-onboarding? (Today: no — single-medication scope.)
- What is the UX for an unrecognized medication name beyond degrading to the manual picker?
- Is there any token revocation/refresh story? ( `/auth/logout` is a no-op; JWTs simply expire after 7 days.)
- The pattern insight is hard-coded to **Wednesday/Thursday** misses — is that intended as the only pattern, or a placeholder for a more general detector?

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*End of overview.*